

Wenjing Lu

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Education

Shanghai Jiao Tong University	Shanghai, China
<i>Ph.D. Candidate in Computer Science</i>	Sept. 2022 – Dec. 2026 (Expected)
- Supervisor: Prof. Yang Yang, Co-Advisor: Prof. Yi Hong - PhD Thesis: Uncertainty Modeling for Medical Image Analysis	
National University of Singapore	Singapore
<i>M.S. in Electrical and Computer Engineering</i>	July 2021 - June 2022
University of Electronic Science and Technology of China	Chengdu, China
<i>B.E. in Electronic and Information Engineering (GPA: 3.94 / 4.00)</i>	Sept. 2017 - June 2021

Research Interests

Principled Uncertainty Quantification and Calibration, Reliable Decision-making and Robustness in Vision-Language Models (VLMs), Label-Efficient Learning via Foundation Model Adaptation (e.g., Segment Anything Model), Out-of-Distribution Generalization, and Uncertainty Modeling in Agentic LLM Systems.

Professional Experiences

RIKEN Center for Advanced Intelligence Project (AIP)	Tokyo, Japan
<i>Student Trainee</i>	Apr. 2025 - Present
- Supervisor: Dr. Qibin Zhao - Research Topic: Uncertainty Quantification in Trustworthy AI	
A*STAR Institute for Infocomm Research (A*STAR I²R)	Singapore
<i>Research Assistant</i>	Aug. 2021 - May 2022
- Supervisor: Prof. Bharadwaj Veeravalli & Dr. Zeng Zeng - Research Topic: Label-Efficient Learning for Biomedical Image Segmentation	

Selected Publications ([Full list](#))

- [1] **Wenjing Lu**; Zerui Tao; Dongping Zhang; Yuning Qiu; Yang Yang; Qibin Zhao. *Calibrating Uncertainty for Zero-Shot Adversarial CLIP*, accepted by **ICML 2026**. [\[Link\]](#)
- [2] **Wenjing Lu**; Yi Hong; Yang Yang. *Harmonizing Generalization and Specialization: Uncertainty-Informed Collaborative Learning for Semi-supervised Medical Image Segmentation*, IEEE Transactions on Medical Imaging (TMI), 2026. [\[Link\]](#)
- [3] **Wenjing Lu**; Yi Hong; Yang Yang. *UP-SAM: Uncertainty-Informed Adaptation of Segment Anything Model for Semi-Supervised Medical Image Segmentation*. In: IEEE International Conference on Bioinformatics and Biomedicine (BIBM), pp. 2256-2261, 2024. [\[Link\]](#)
- [4] **Wenjing Lu**; Jiahao Lei; Peng Qiu; Rui Sheng; Jinhua Zhou; Xinwu Lu; Yang Yang. *Upcol: Uncertainty-Informed Prototype Consistency Learning for Semi-supervised Medical Image Segmentation*. In: International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), pp. 662–672, 2023. *Early Acceptance*. [\[Link\]](#)
- [5] Zhiyuan Ding*; **Wenjing Lu*** (Equal Contribution); Ling Wang; Xiangzhu Zeng; Tong Zhao; Xu Tian; Zeng Wang; Yan Liu. *Voxel-Level fMRI Analysis by Representation Learning and Deep Clustering for Alzheimer's Disease*. In: 2023 IEEE 20th International Symposium on Biomedical Imaging (ISBI), pp. 1–5, 2023. [\[Link\]](#)
- [6] Ziyuan Zhao*; **Wenjing Lu*** (Equal Contribution); Zeng Zeng; Kaixin Xu; Bharadwaj Veeravalli; Cuntai Guan. *Self-supervised assisted active learning for skin lesion segmentation*. In: 2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), pp. 5043–5046, 2022. [\[Link\]](#)

Services

- **Professional Leadership & Reviewing:**

- Area Chair: ICLR 2026 Workshop DeLTa, ICML 2026 Workshop FoGen
- Conference Reviewer: ICML (**Gold Reviewer**), NeurIPS, MICCAI
- Journal Reviewer: Engineering Applications of Artificial Intelligence (EAAI)

- **Teaching Assistantship:**

- CS7336: Machine Learning (Graduate Level), SJTU, Spring 2024

Research Experience

Calibrating Uncertainty for Zero-Shot Adversarial CLIP

RIKEN Center for Advanced Intelligence Project (AIP)

Tokyo, Japan

Student Trainee, Supervisor: Dr. Qibin Zhao

Apr. 2025 - Present

- Identified an *uncertainty inversion* phenomenon in zero-shot adversarial CLIP, where adversarial perturbations reduce predictive entropy and reveal a key calibration gap for reliable robustness.
- Formulated a *principled evidential view* of the contrastive Vision-Language Models by interpreting normalized image-text logits as Dirichlet concentration parameters, enabling closed-form decomposition of aleatoric and epistemic uncertainty while preserving semantic logit geometry.
- Developed UCAT [\[Link\]](#), an **uncertainty-calibrated adversarial fine-tuning objective** that aligns clean and adversarial Dirichlet distributions, jointly preserving semantic structure and evidence strength through KL-based evidential matching with a text-anchor constraint in a single pass, thereby improving the reliability of zero-shot predictions under perturbations.

Uncertainty Estimation and Modeling for Semi-supervised Medical Image Analysis

Shanghai Jiao Tong University

Shanghai, China

Ph.D. Student, Supervisor: Prof. Yang Yang, Co-Advisor: Prof. Yi Hong

Sept. 2022 - Present

- Developed UPCoL [\[Link\]](#), an **uncertainty-informed prototype consistency framework** that updates and fuses labeled prototypes and reliability-filtered unlabeled prototypes in the latent space, and applies non-parametric metric learning to enforce voxel-wise prototype consistency with pseudo-labels, yielding more stable class-center representations and improving reliability in unlabeled regions.
- Designed UP-SAM [\[Link\]](#), an **uncertainty-informed adaptation of Segment Anything Model** that reduces the *epistemic gap* under scarce labels via multi-head supervised learning and partial decoder fine-tuning, while leveraging unsupervised stochastic modeling on unlabeled data to capture *aleatoric uncertainty* and regularize domain-specific predictions with foundation-model guidance.
- Proposed UnCoL [\[Link\]](#), an **Uncertainty-Informed Collaborative Learning** framework that (i) performs *dual-path knowledge distillation* to extract generalized knowledge from the image encoder and prompt-conditioned mask decoder while enabling prompt-free learning, and (ii) uses *uncertainty-guided pseudo-labeling* to selectively trust reliable regions and coordinate the generalized teacher (foundation model) with the specialized teacher (EMA model).

Awards

- Gold Reviewer, ICML 2026 Forty-Third International Conference on Machine Learning
- Student Travel Award, 2024 IEEE International Conference on Bioinformatics and Biomedicine
- Winner of Artificial Intelligence Research Student Conference in Singapore (**2nd place**)
- Outstanding Graduate of University of Electronic Science and Technology of China

Skills

Programming: Python, Matlab.

Languages: Chinese (Native), English (Proficient), Japanese (Learning)